BSTA 551: Statistical Inference

Review: Point Estimation (Lessons 1-5)

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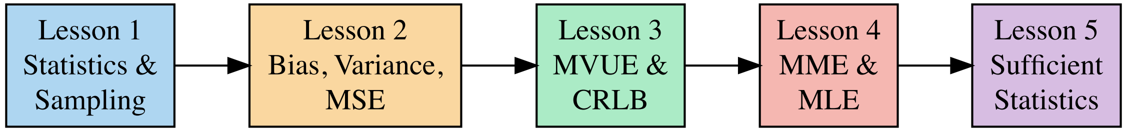
Review: The Big Picture of Point Estimation

Where We've Been: The Journey So Far



Today: Tie it all together and review key concepts.

The Central Problem of Statistical Inference

We observe data $X_1, X_2, \ldots, X_n$ from some distribution with unknown parameter θ. Usually they are identically and independently distributed (i.i.d.), so they make a "random sample."

The Question: How do we learn about θ from the data?

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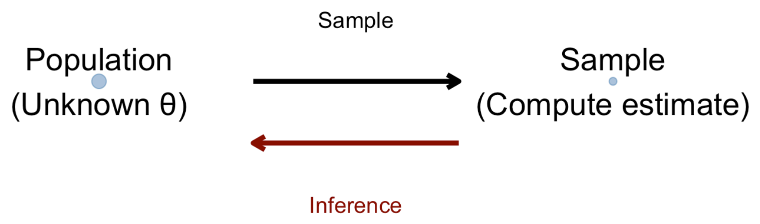

The Two Key Questions in Point Estimation

Question 1: How do we **construct** estimators?

- Method of Moments (MME)
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- Bias (accuracy)
- Variance (precision)
- Mean Squared Error (overall quality)
- Efficiency (comparison to optimal)

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Question 3: How do we find the **best** estimator?

- MVUE: Minimum Variance Unbiased Estimator
- Sufficient statistics: Data reduction without information loss

Part 1: Properties of Estimators

The Fundamental Tradeoff: Bias vs. Variance

Key Definitions

- **Bias:** $\mathrm{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$ (systematic error)
- **Variance:** $\mathrm{Var}(\hat{\theta}) = E[(\hat{\theta} - E(\hat{\theta}))^2]$ (random error)
- **MSE:** $\mathrm{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}^2$

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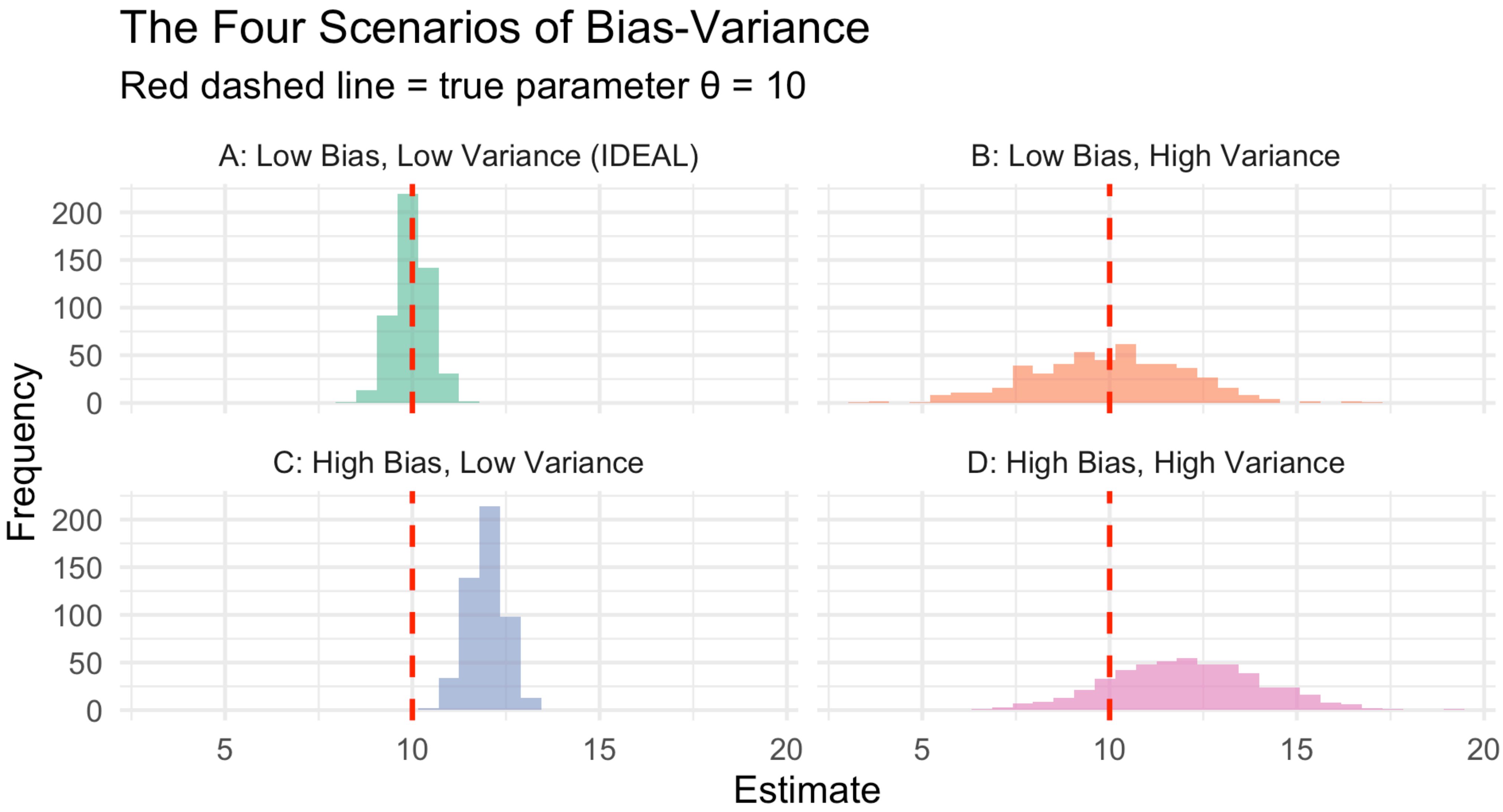
The key insight: MSE = Variance + Bias^2

This means a **biased** estimator can be better than an unbiased one if its variance is low enough!

Visual Intuition: Bias and Variance



Medical Example: Estimating Treatment Effect

Scenario: Estimating the true cure rate p of a new cancer treatment.

Two estimators:

- **Standard:** $\hat{p}_1 = X/n$ (sample proportion, unbiased)
- **Add-Two:** $\hat{p}_2 = (X+2)/(n+4)$ (biased toward 0.5)

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Understanding the "Add-Two" Estimator: The Intuition

What does it do? It acts like you observed 2 extra successes and 2 extra failures before seeing the data.

$$\hat{p}_2 = \frac{X+2}{n+4} = \frac{\text{successes}+2}{\text{trials}+4}$$

Why +2 in numerator and +4 in denominator?

- Adding 2 successes and 2 failures = adding 4 total "pseudo-observations"
- This "pulls" the estimate toward $2/4 = 0.5$ (the middle)
- The pull is stronger when n is small, weaker when n is large

Why Does the Add-Two Estimator Help?

The Problem with Standard Estimator:

- If you observe 0 successes in 10 trials, $\hat{p}_1 = 0/10 = 0$
- But is the true p really exactly 0? Probably not!
- Small samples give extreme estimates that are likely wrong

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How Add-Two Helps:

- With 0 successes in 10 trials: $\hat{p}_2 = (0+2)/(10+4) = 2/14 \approx 0.14$
- This is more “conservative” — it hedges against extreme estimates
- **Shrinkage toward 0.5** reduces variance at the cost of small bias

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> **Bayesian Connection**
>
> The add-two estimator is the **posterior mean** under a Uniform(0,1) prior on p. It incorporates prior belief that extreme values of p (near 0 or 1) are less likely.

Comparing the Two Estimators: Simulation

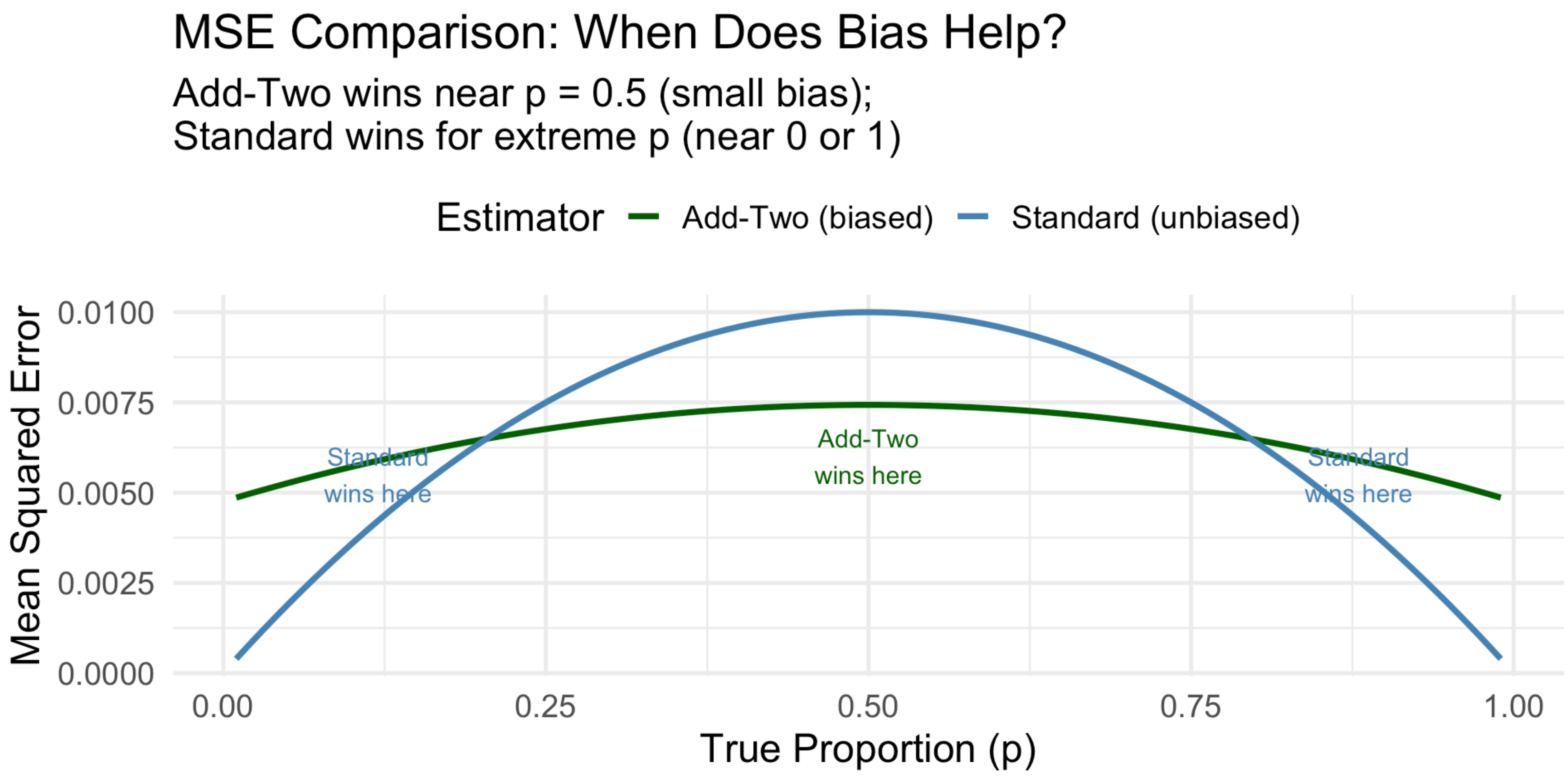
For $p = 0.2$ (a fairly extreme value), let's compare the two estimators:

▶ Code

```
# A tibble: 4 × 3
  Metric   Standard `Add-Two`
  <chr>       <dbl>     <dbl>
1 Mean      0.199     0.241
2 Bias     -0.00101   0.0405
3 Variance  0.0065    0.00483
4 MSE       0.0065    0.00647
```

At $p = 0.2$, the MSEs are similar — we're in the "crossover" region where neither dominates!

When Does Add-Two Win? MSE Comparison



Key insight: The add-two estimator wins **near** $p = 0.5$ where its bias is small, but loses at **extreme** p **values** where shrinkage toward 0.5 hurts!

What Properties Do We Want in an Estimator?

Property	Definition	Why It Matters
Unbiased	$E(\hat{\theta}) = \theta$	No systematic error
Low Variance	$\mathrm{Var}(\hat{\theta})$ small	Precise estimates
Consistent	$\hat{\theta} \xrightarrow{P} \theta$	Works with enough data
Efficient	Achieves CRLB	Best among unbiased

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The Hierarchy of Goals

1. **Consistency** is the bare minimum
2. **Unbiasedness** is nice but not essential as long as bias shrinks as $n \to \infty$
3. **Efficiency** (low variance) is what really matters for precision

Part 2: Constructing Estimators

Two Systematic Methods: MME and MLE

Method of Moments (MME):

- Match population moments to sample moments
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Maximum Likelihood (MLE):

- Find θ that maximizes $P(\mathrm{data}|\theta)$
- More complex, but has optimal properties
- Asymptotically efficient, unbiased, Normally distributed!

The MME Recipe

Method of Moments: The Algorithm

1. Express the first k population moments as functions of θ: $E(X), E(X^2), \ldots$
2. Set each equal to its sample counterpart: $\bar{X}, \frac{1}{n}\sum X_i^2, \ldots$
3. Solve for θ

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Example: Exponential(λ)

- Population moment: $E(X) = 1/\lambda$
- Set equal: $1/\lambda = \bar{X}$
- Solve: $\hat{\lambda}_{MME} = 1/\bar{X}$

The MLE Recipe

Maximum Likelihood: The Algorithm

1. Write the likelihood: $L(\theta) = \prod_{i=1}^{n} f(x_i; \theta)$
2. Take the log: $\ell(\theta) = \sum_{i=1}^{n} \ln f(x_i; \theta)$
3. Differentiate and set to zero: $\ell'(\theta) = 0$
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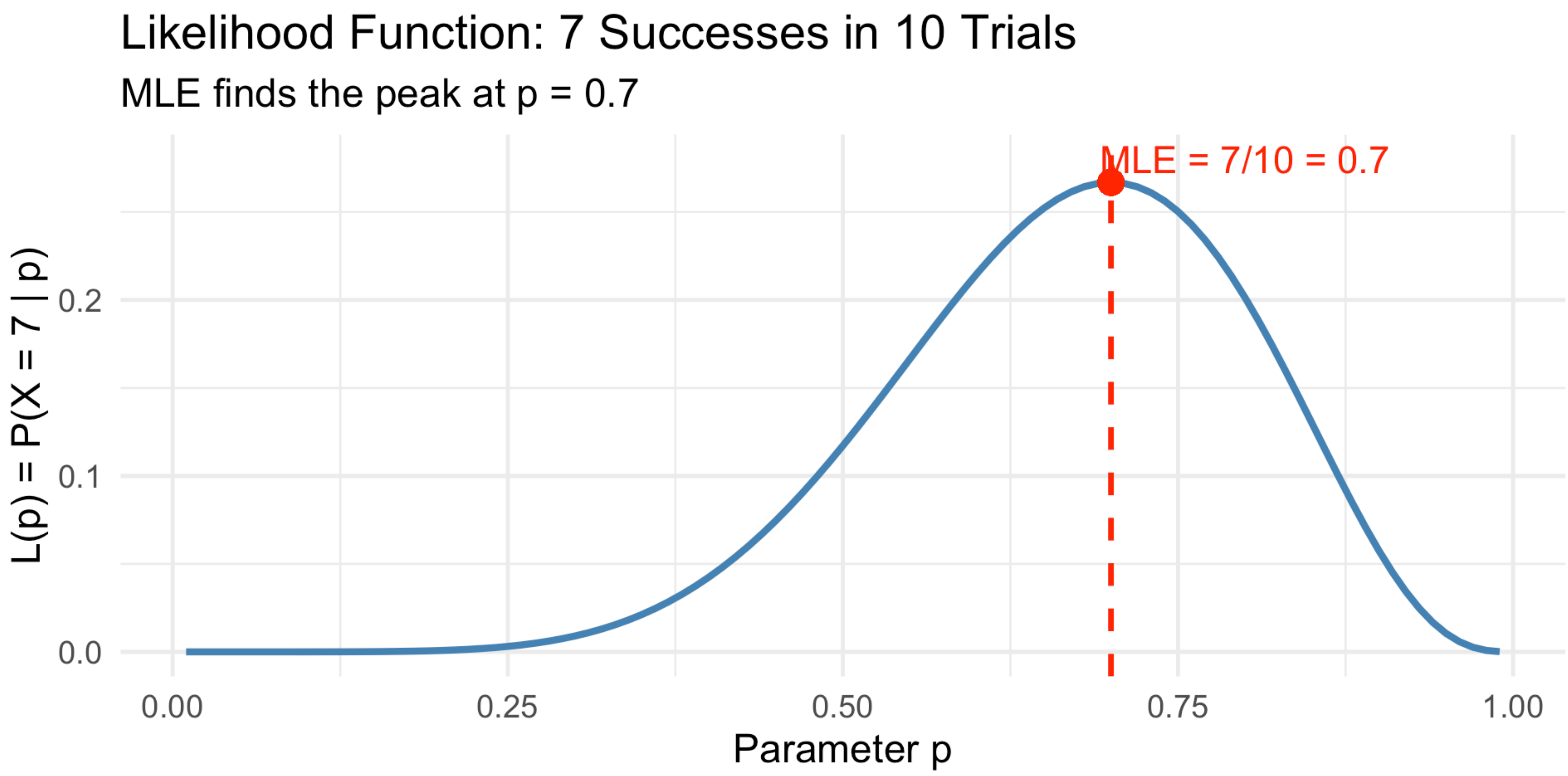
Intuition: The MLE asks "What parameter value makes our observed data most probable?"

Visual Intuition: What Does the MLE Do?



The MLE finds where the likelihood is **highest** — the most "likely" parameter value.

When Do MME and MLE Agree?

Distribution	MLE	MME	Same?
Bernoulli(p)	$\bar{X}$	$\bar{X}$	✓
Normal(μ, known σ^2)	$\bar{X}$	$\bar{X}$	✓
Exponential(λ)	$1/\bar{X}$	$1/\bar{X}$	✓
Poisson(μ)	$\bar{X}$	$\bar{X}$	✓
Uniform[0, θ]	$\max(X_i)$	$2\bar{X}$	✗

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Key Insight

When they differ, the MLE is usually preferred (especially for large n) because it's asymptotically efficient.

MLE Properties: Why It's Often the Best Choice

Three Key Properties of MLEs (Devore 7.2)

1. **Invariance:** If $\hat{\theta}$ is the MLE of θ, then $g(\hat{\theta})$ is the MLE of $g(\theta)$
2. **Consistency:** $\hat{\theta}_{MLE} \xrightarrow{P} \theta$ as $n \to \infty$
3. **Asymptotic Efficiency:** For large n, $\hat{\theta}_{MLE} \dot{\sim} N\left(\theta, \frac{1}{nI(\theta)}\right)$

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In plain English:

- You can transform MLEs freely (invariance)
- MLEs "work" given enough data (consistency)
- MLEs are approximately the **best** for large samples (efficiency)

Consistency: What Does It Really Mean?

Definition: Consistency (Devore 7.1, Chihara 6.3.4)

An estimator $\hat{\theta}_n$ is **consistent** if:

$$P(|\hat{\theta}_n - \theta| < \epsilon) \to 1 \quad \text{as } n \to \infty$$

for any $\epsilon > 0$, no matter how small.

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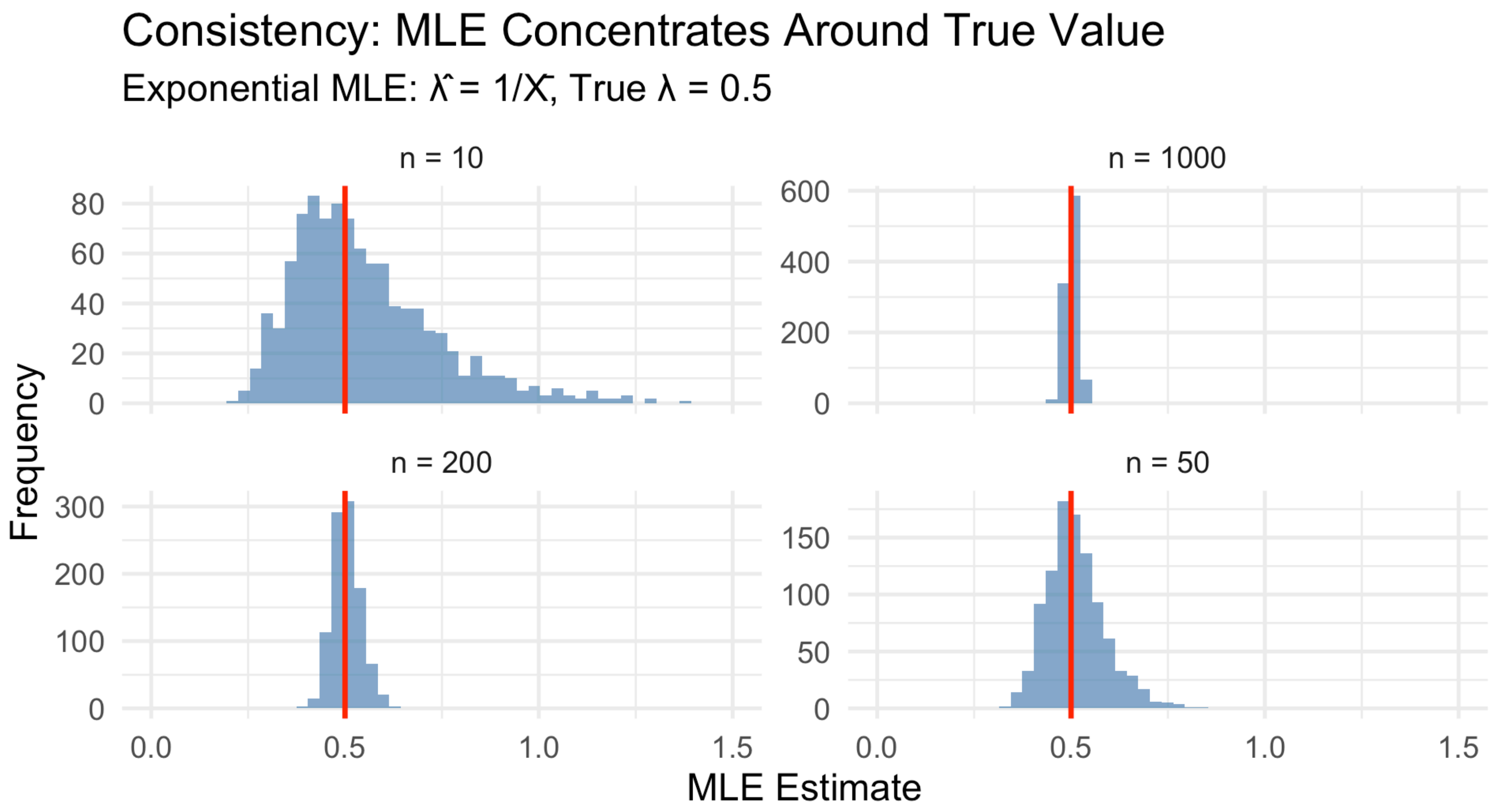
This requires:

- Variance $\to 0$ (estimates get more precise)
- Bias $\to 0$ (estimates get more accurate)

Visualizing Consistency



As n increases, the distribution "shrinks" around the truth!

Part 3: Finding the Best Estimator

The MVUE: Minimum Variance Unbiased Estimator

Definition (Devore 7.1)

The **MVUE** of θ is the estimator that:

1. Is unbiased: $E(\hat{\theta}) = \theta$
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Why not just minimize MSE?

- MSE depends on the true θ (which we don't know!)
- A biased estimator might be better at some θ values, worse at others
- MVUE provides a clean, universal optimality criterion

The Cramér-Rao Lower Bound (CRLB)

Theorem: CRLB (Devore 7.4)

For any unbiased estimator T of θ:

$$\text{Var}(T) \geq \frac{1}{nI(\theta)}$$

where $I(\theta)$ is the **Fisher Information**:

$$I(\theta) = E\left[-\frac{\partial^2}{\partial\theta^2}\ln f(X;\theta)\right]$$

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Interpretation:

- The CRLB is a **floor** — no unbiased estimator can do better
- If an estimator achieves the CRLB, it's the **MVUE**
- Fisher Information measures the "curvature" of the log-likelihood

Fisher Information: The Intuition

Key Insight: Fisher Information measures how much the log-likelihood changes as θ changes.

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High Fisher Information:

- Sharp, peaked likelihood → easy to pinpoint θ
- Small variance possible

Low Fisher Information:

- Flat likelihood → hard to distinguish θ values
- Large variance unavoidable

Efficiency: How Close to Optimal?

Definition: Efficiency

An unbiased estimator T is **efficient** if:

$$\text{Var}(T) = \frac{1}{nI(\theta)} = \text{CRLB}$$

More generally, the **efficiency** of T is:

$$\text{Efficiency}(T) = \frac{\text{CRLB}}{\text{Var}(T)} \leq 1$$

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Example: Sample median for normal data

- Variance ≈ $1.57 \cdot \sigma^2/n$
- CRLB = σ^2/n
- Efficiency = $1/1.57 \approx 0.64$ (64% efficient)

Key Results: Efficient Estimators

Distribution	Parameter	MVUE	CRLB	Efficient?
Bernoulli(p)	p	$\bar{X}$	$\frac{p(1-p)}{n}$	✓
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Note on Exponential(λ): Why No Efficient Unbiased Estimator Exists

For the exponential distribution with **rate parameter** λ:

- **MLE:** $\hat{\lambda}_{MLE} = 1/\bar{X}$ is **biased** – $E(1/\bar{X}) = \frac{n\lambda}{n-1} \neq \lambda$
- **MVUE:** The bias-corrected estimator $\hat{\lambda}_{MVUE} = \frac{n-1}{\sum X_i} = \frac{n-1}{n} \cdot \frac{1}{\bar{X}}$
- **Variance of MVUE:** $\mathrm{Var}(\hat{\lambda}_{MVUE}) = \frac{\lambda^2}{n-2}$ (for $n > 2$)
- **CRLB:** $\frac{\lambda^2}{n}$

Since $\frac{\lambda^2}{n-2} > \frac{\lambda^2}{n}$, the MVUE does **not** achieve the CRLB. No unbiased estimator can!

Alternatively: For the **mean parameter** $\mu = 1/\lambda$, the sample mean $\bar{X}$ **is** the efficient MVUE.

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Note on Uniform[0, θ]

Part 4: Sufficient Statistics — The Key to Everything

Why Sufficiency Matters: The Central Concept

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Answer: Sufficient statistics answer a fundamental question:

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> **What is the minimum amount of information we need to keep from our data to estimate θ?**

If a statistic T is sufficient, then:

- We can throw away the raw data and just keep T
- We lose **no information** about θ
- All optimal estimators will be functions of T

Intuition: What Does "Sufficient" Really Mean?

Scenario: You observe n coin flips and want to estimate p (probability of heads).

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Question: Does knowing the exact sequence (HTHHTHTTHH) tell you more about p than just knowing “6 out of 10”?

Answer: No! The total number of heads captures all the information about p.

This is why $T = \sum X_i$ is **sufficient** for p in the Bernoulli model.

Formal Definition of Sufficiency

Definition (Devore 7.3)

A statistic $T = t(X_1, \ldots, X_n)$ is **sufficient for** θ if:

$$P(X_1 = x_1, \ldots, X_n = x_n \mid T = t) \text{ does not depend on } \theta$$

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In words: Given the value of T, the conditional distribution of the data doesn't involve θ.

Intuition: Once you know T, learning the raw data tells you nothing new about θ.

The Neyman Factorization Theorem

Computing conditional distributions is hard. The **factorization theorem** gives us a shortcut!

Theorem (Neyman-Fisher Factorization)

T is sufficient for θ if and only if the joint pdf/pmf can be written as:

$$f(x_1, \ldots, x_n; \theta) = g(T; \theta) \cdot h(x_1, \ldots, x_n)$$

where:

- g involves θ and depends on data **only through** T
- h depends on data but **not on** θ

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where:

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Strategy: Look for parts involving θ and group them with a function of the data.

Example: Finding the Sufficient Statistic for Poisson

Setup: $X_1, \ldots, X_n \overset{iid}{\sim} \text{Poisson}(\mu)$

Joint pmf:

$$f(x_1, \ldots, x_n; \mu) = \prod_{i=1}^{n} \frac{e^{-\mu}\mu^{x_i}}{x_i!} = \frac{e^{-n\mu}\mu^{\sum x_i}}{\prod x_i!}$$

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Factor it:

$$= \underbrace{e^{-n\mu}\mu^{T}}_{g(T;\mu)} \cdot \underbrace{\frac{1}{\prod x_i!}}_{h(x_1,\ldots,x_n)}$$

where $T = \sum x_i$.

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where $T = \sum x_i$.

Result

$T = \sum_{i=1}^{n} X_i$ is sufficient for μ.

Example: Uniform[0, θ] — A Surprising Result

Setup: $X_1, \ldots, X_n \overset{iid}{\sim} \text{Uniform}[0, \theta]$

Joint pdf:

$$f(x_1, \ldots, x_n; \theta) = \frac{1}{\theta^n} \cdot I(0 \leq \min(x_i)) \cdot I(\max(x_i) \leq \theta)$$

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Setup: $X_1, \ldots, X_n \overset{iid}{\sim} \text{Uniform}[0, \theta]$

Joint pdf:

$$f(x_1, \ldots, x_n; \theta) = \frac{1}{\theta^n} \cdot I(0 \leq \min(x_i)) \cdot I(\max(x_i) \leq \theta)$$

Factor it:

$$= \underbrace{\frac{1}{\theta^n} I(\max(x_i) \leq \theta)}_{g(T;\theta)} \cdot \underbrace{I(\min(x_i) \geq 0)}_{h(\mathbf{x})}$$

Example: Uniform[0, θ] — A Surprising Result

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Joint pdf:

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Key Result

$T = \max(X_i)$ is sufficient for θ.

Notice: $\bar{X}$ is **NOT** sufficient for the uniform distribution!

Why Isn't $\bar{X}$ Sufficient for Uniform?

```r
# Two datasets with same mean but different information about θ
data_A <- c(2, 4, 6, 8)   # mean = 5, max = 8
data_B <- c(1, 4, 5, 10)  # mean = 5, max = 10

cat("Dataset A: mean =", mean(data_A), ", max =", max(data_A), "\n")
```

```
Dataset A: mean = 5 , max = 8
```

```r
cat("Dataset B: mean =", mean(data_B), ", max =", max(data_B), "\n")
```

```
Dataset B: mean = 5 , max = 10
```

Why Isn't $\bar{X}$ Sufficient for Uniform?

```r
# Two datasets with same mean but different information about θ
data_A <- c(2, 4, 6, 8)   # mean = 5, max = 8
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```

```
Dataset A: mean = 5 , max = 8
```

```r
cat("Dataset B: mean =", mean(data_B), ", max =", max(data_B), "\n")
```

```
Dataset B: mean = 5 , max = 10
```

Both have $\bar{X} = 5$, but:

- Dataset A tells us $\theta \geq 8$
- Dataset B tells us $\theta \geq 10$

Knowing only $\bar{X}$ loses critical information about θ!

Sufficient Statistics: Summary Table

Distribution	Parameter(s)	Sufficient Statistic(s)
Bernoulli(p)	p	$\sum X_i$
Poisson(μ)	μ	$\sum X_i$
Exponential(λ)	λ	$\sum X_i$
Normal(μ, known σ^2)	μ	$\bar{X}$
Normal(μ, σ^2)	(μ, σ^2)	$(\bar{X}, S^2)$ or $(\sum X_i, \sum X_i^2)$
Uniform[0, θ]	θ	$\max(X_i)$

Why Sufficiency Matters: The Big Picture

Three Key Connections

1. **MLEs are functions of sufficient statistics**
   - You don't lose anything by working with T instead of raw data
2. **MVUEs are functions of sufficient statistics** (Rao-Blackwell)
   - The best unbiased estimator depends only on T
3. **If T is complete and sufficient, any unbiased function of T is the MVUE** (Lehmann-Scheffé)
   - Finding the MVUE becomes much easier!

The Rao-Blackwell Theorem: Improving Estimators

Theorem (Rao-Blackwell)

If U is any unbiased estimator of $\tau(\theta)$ and T is sufficient for θ, then:

$$U^* = E(U \mid T)$$

is also unbiased, and $\mathrm{Var}(U^*) \leq \mathrm{Var}(U)$.

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In plain English: Conditioning on a sufficient statistic **never hurts** and usually **helps**.

Practical implication: If your estimator doesn't use all the information in T, you can always improve it!

Medical Example: Rao-Blackwell in Action

Problem: Estimate $e^{-\mu} = P(\text{no defects})$ for Poisson(μ) data.

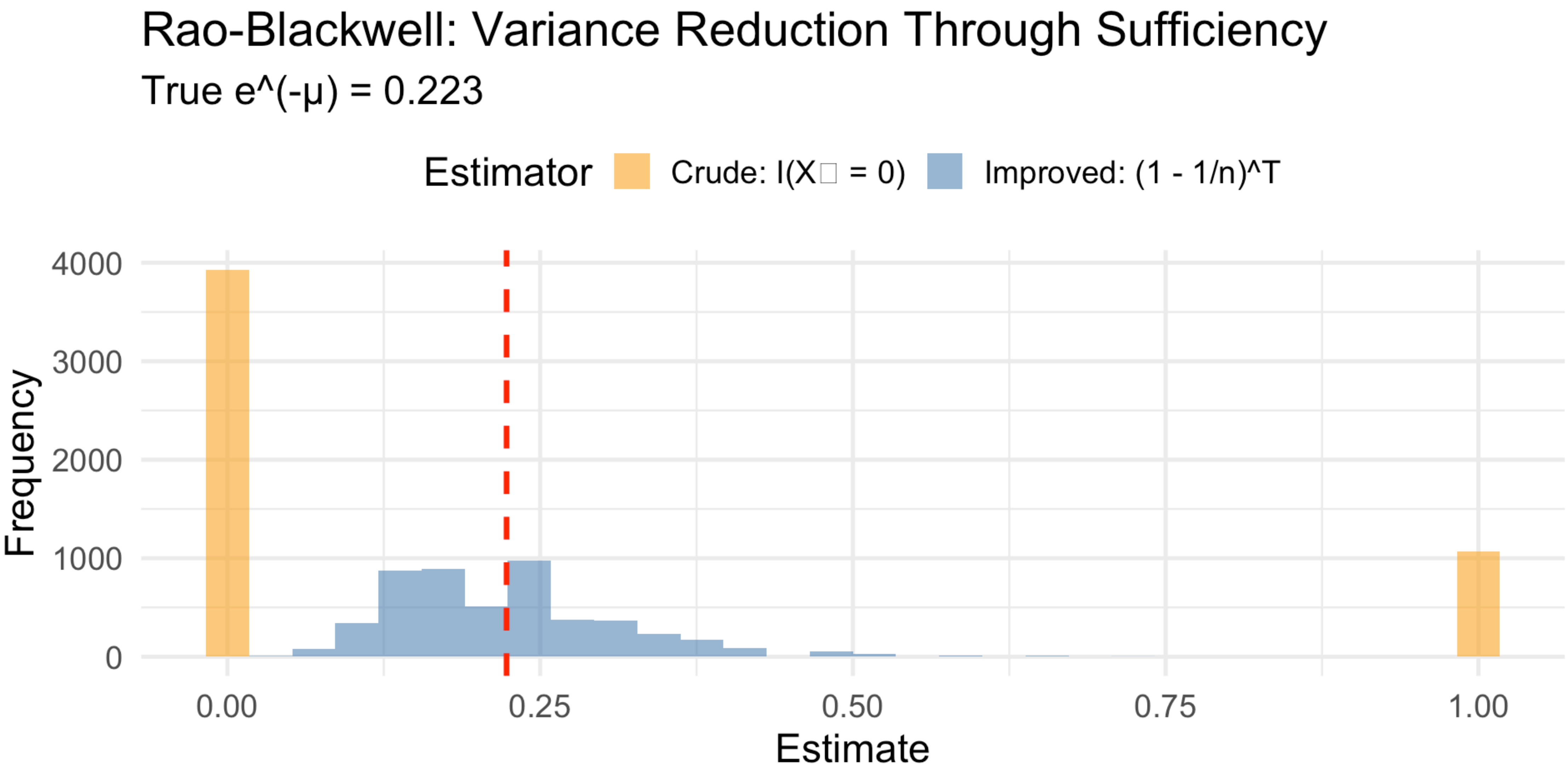
Crude estimator: $U = I(X_1 = 0)$ (is the first observation zero?)

▶ Code

```
# A tibble: 5 × 2
  name              value
  <chr>             <dbl>
1 True e^(-μ)      0.223
2 Crude: E[U]      0.214
3 Crude: Var       0.168
4 Improved: E[U*]  0.223
5 Improved: Var    0.0082
```

The Rao-Blackwell estimator has **much lower variance**!

Visualization: Rao-Blackwell Improvement



Completeness: The Final Piece

Definition: Complete Statistic

T is **complete** if for any function g:

$$E_\theta[g(T)] = 0 \text{ for all } \theta \implies P_\theta(g(T) = 0) = 1 \text{ for all } \theta$$

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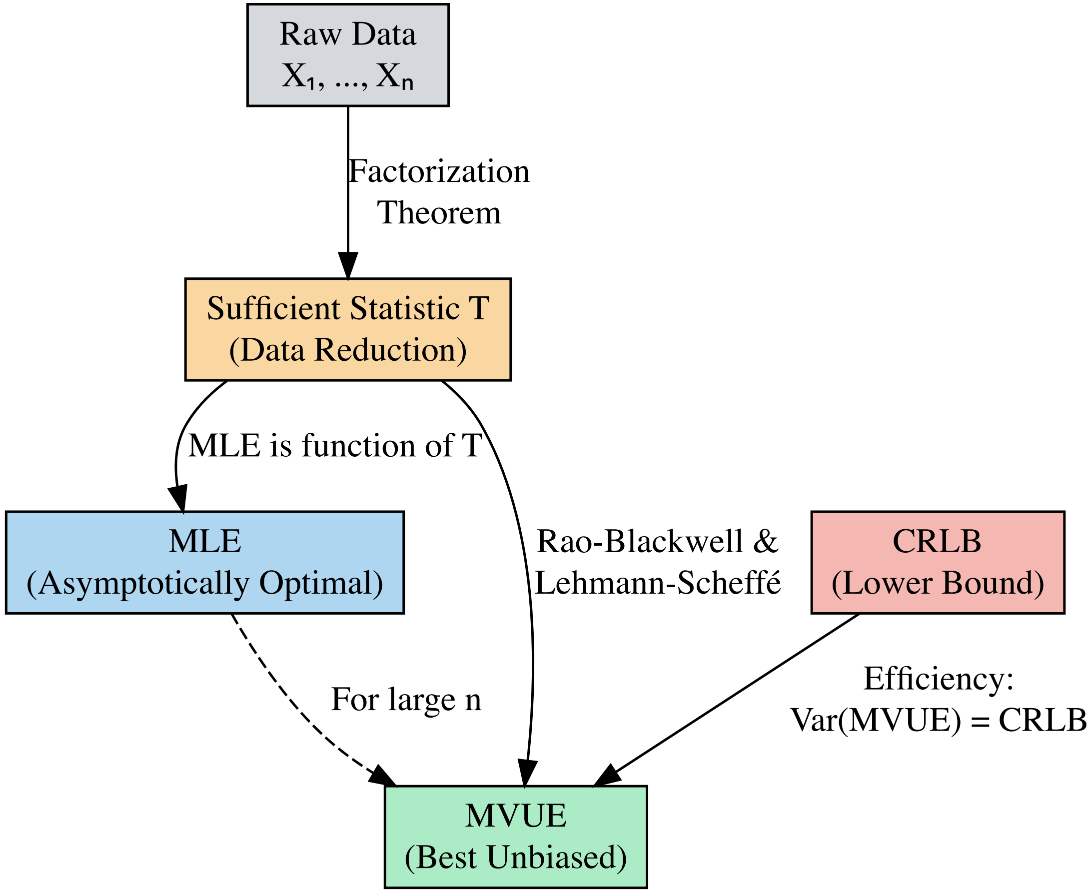
Lehmann-Scheffé Theorem

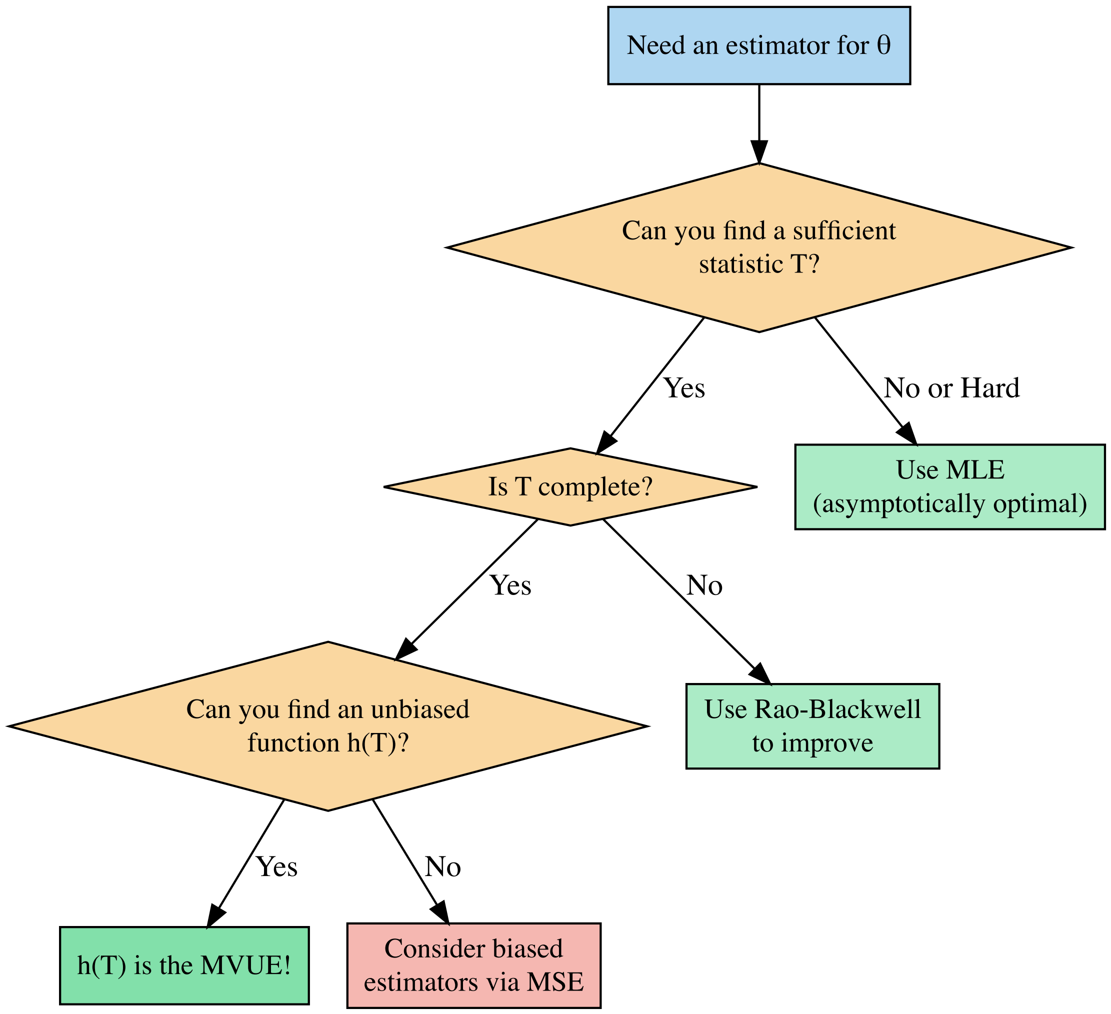
If T is **complete** and **sufficient**, and $h(T)$ is unbiased for $\tau(\theta)$, then $h(T)$ is the **unique MVUE** of $\tau(\theta)$.

Part 5: Putting It All Together

The Complete Picture: How Everything Connects



Decision Tree: Finding the Best Estimator



(Note: Claude AI put these flow charts together for me, thanks Claude!)

Summary: Key Formulas to Remember

Concept	Formula
Bias	$E(\hat{\theta}) - \theta$
MSE	$\text{Var}(\hat{\theta}) + \text{Bias}^2$
Standard Error	$\sqrt{\text{Var}(\hat{\theta})}$
Fisher Information	$I(\theta) = E[-\ell''(\theta)]$
CRLB	$\frac{1}{nI(\theta)}$
Efficiency	$\frac{\text{CRLB}}{\text{Var}(\hat{\theta})}$
Consistency	$\text{MSE} \to 0$ as $n \to \infty$

Summary: Key Results by Distribution

Distribution	MLE	MME	Sufficient Stat	MVUE of Mean
Bernoulli(p)	$\bar{X}$	$\bar{X}$	$\sum X_i$	$\bar{X}$
Poisson(μ)	$\bar{X}$	$\bar{X}$	$\sum X_i$	$\bar{X}$
Exponential(λ)	$1/\bar{X}$	$1/\bar{X}$	$\sum X_i$	$\bar{X}$ (for mean)
Normal(μ, σ^2)	$\bar{X}$, $\frac{1}{n}\sum(X_i - \bar{X})^2$	Same	$(\bar{X}, S^2)$	$\bar{X}$
Uniform[0, θ]	$\max(X_i)$	$2\bar{X}$	$\max(X_i)$	$\frac{n+1}{n}\max(X_i)$

Common Misconceptions to Avoid

Pitfall 1: "Unbiased is always better"

Reality: A biased estimator with low variance can have lower MSE!

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Reality: MLE is a **function of** sufficient statistics, not the sufficient statistic itself.

Common Misconceptions to Avoid

Pitfall 1: "Unbiased is always better"

Reality: A biased estimator with low variance can have lower MSE!

Pitfall 2: "MLE = Sufficient Statistic"

Reality: MLE is a **function of** sufficient statistics, not the sufficient statistic itself.

Pitfall 3: "$\bar{X}$ is always sufficient"

Reality: $\bar{X}$ is sufficient for location parameters (normal, exponential) but NOT for scale parameters like in Uniform$[0, \theta]$.

References

- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Chapters 6-7.
- Chihara and Hesterberg. *Mathematical Statistics with Resampling and R* (Wiley). Chapter 6.

Questions?

Thank you!